

Satellite-based Estimation of Above Ground Carbon Stock Estimation for Rubber Plantation in Tembir Salatiga Central Java

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Abstract— The concentration of CO₂ content in the atmosphere needs to be reduced in order to reduce the greenhouse effect. The effective way of CO₂ absorption in the air is through photosynthesis performed by vegetation. Through photosynthesis, the CO₂ absorbed by vegetation and converted into organic carbon stored in biomass. This research was conducted in a rubber plantation at Tembir Salatiga Central Java. This research aims to determine aboveground carbon stocks that exist in the research area by utilizing remote sensing data. This research uses the image generated by the TERRA satellite sensors, that produced by the project's partnership between the Japan - USA to resolve the problems concerned with natural resources and environmental issue that is the image of the ASTER (Advanced Spaceborne Thermal Emission and Reflection Radiometer). The method used in this research is remote sensing data extraction and statistical analysis. The data is the correlation between biomass data at each sample plots with vegetation index value (MSAVI). The value of R is the correlation between those two data that is used to find out how big the relations between both variables that be correlated. the results of this research are the relationship between vegetation index transform (MSAVI) with biomass content obtained from allometric calculation. The relationship of these two variables indicated by the value of the linear regression which of these research produced a regression between MSAVI and biomass content is $y = 1683.9x - 423.87$ with R^2 0.681 (68%). From these results can be seen that the content of the biomass resulting from the extraction of remote sensing data able to represent 68% of the biomass conditions contained in the study sites. The research results show that carbon stocks contained in the research sites were 875227.25 TonC.

Keywords— Biomass; Remote Sensing Data; MSAVI; Carbon Stock

I. INTRODUCTION

Indonesia is located on a tropical region is a great place for growing plants. As an archipelago country, Indonesia has a lot of potential for forest resources. Plant rubber plantation is one of commodity that has an important role in the life of

the Indonesian economy and it has the world's largest rubber plantation [1].

The issue of global warming is still a major threat to human life, i.e. in the form of impaired health, food security, and environmental degradation. Changes in environmental conditions occur, the increase of the temperature in the atmosphere caused by rising concentrations of CO₂ in the air. In dealing with issues of environmental change, the most important thing is the lower content of CO₂ in the atmosphere. Various things done in his efforts, namely in improving low emissions technology projects, such as minimal emissions of vehicles, or the manufacture of household appliances based eco-friendly [1].

However the development of these projects could not be carried out in all countries, of course, every country has a different policy in addressing the issue of global warming. The Effort to decreased content of CO₂ in the atmosphere, it also needs to be balanced with an increased absorption of CO₂ content in the atmosphere. How effective absorption of CO₂ content in the air through photosynthesis is performed by vegetation. Through photosynthesis, CO₂ is absorbed by vegetation and converted into organic carbon that stored in the form of biomass [12], [17]. This research aims to determine the utilization of remote sensing technology for the estimation of surface carbon stocks on vegetation objects. The utilization of remote sensing data is expected to facilitate the acquisition of data processing of remote sensing applications with wide area coverage and accurate data.

II. METHOD AND SYSTEM DESIGN

Administratively rubber plant that became object of study of this research is located in area Tembir Salatiga Central Java province with the coordinates 444,800.391 mT – 9,194,335.339 mU. Research method is steps which must be performed in collection, processing and analysis data to describe the problem-solving research. The methods used in this study was the combined analysis of remote sensing imagery, geographic information systems (GIS), and field

measurement. Sampling approaches intended to take information that cannot be scaled back directly using remote sensing data such as the diameter of the trunk, canopy cover width, etc. calculation of carbon stock is taken from the results of sampling data in field by using allometric.

The vegetation index is a number stating the magnitude of value or the height of a phenomenon associated with vegetation characteristic. The transformation of the vegetation index can be defined as a pixel value changing on the transformations of the image, thus generating a new image with pixel values represent variations of the phenomenon associated with vegetation density, biomass, chlorophyll content, etc [4], [14].

Information retrieval techniques of remote sensing data is used in to retrieve information [2], [5] about vegetation such as density and the characteristics of the vegetation that became the object of research studies [6]. It will then generate a data unit which covers the entirety information of respective data. These data are then used as the basis for going to take information that couldn't be extract using remote sensing data with field work [16]. The information that taken when the field work is information on the diameter of the tree in the first branch, and high of trees which were then computed into biomass with allometric equations or biomass calculations. The biomass data taken using a unit pixel of remote sensing image data [11], [13].

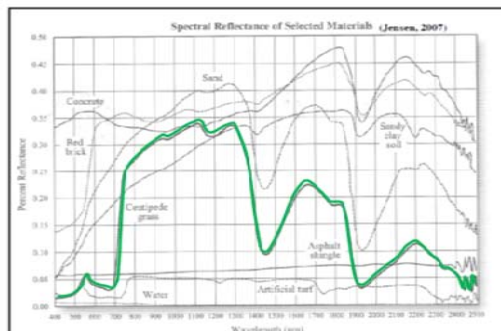


Fig. 1. The spectral pattern reflectance of vegetation (green), [7], [10].

A. Data Acquisition

Remote sensing data after radiometric and geometric correction then processed using vegetation index to determine the density of the vegetation. The density of the vegetation is the first stage of interaction between waves that are used in remote sensing with the object of vegetation. The transformation of the vegetation index used is MSAVI (Modified Soil Adjusted Vegetation Index) with a transformation formula:

$$\text{MSAVI} = \frac{\{2(\text{NIR}) + 1 - \sqrt{\{2(\text{NIR}) + 1\}^2 - 8\{(\text{NIR}) - \text{Red}\}}\}}{2} \quad (1)$$

The fieldwork was conducted to retrieve information about stem diameter, canopy width, and height of branch-free trunk based on the transformation. This measurement is divided by the size of the existing vegetation on each plot samples. Installation of sample plots used for field work adapted to the spatial resolution of remote sensing data are used. The remote sensing data are used has a spatial resolution of 15 meters for multispectral band, so samples plot are used for field work is 45 meters x 45 meters.

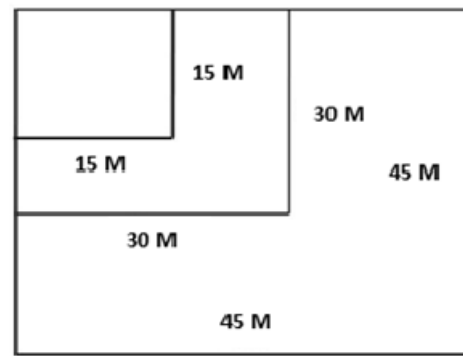


Fig. 2. Sample plot field measurements

The next step after collection of field data the next stage is the post-field analysis that is processing the data obtained during field work and data extraction from remote sensing images using statistical analysis. The analysis of these statistical data is the correlation and regression analysis. correlation is used to determine the direction and the strong association between the Y variable which is the width of canopy cover in the capture of data on each samples in the field with the X variable which is the value of pixel brightness which obtained from the transformation of vegetation index from ASTER data, while the regression analysis is used to determine the influence of the caused by changes in each unit of the Y variable. The correlation coefficient (r) is used to determine the association between the two variables. Strong and weak relationship between the two variables is affected by the same changes between the variables X and variables Y [17], [15].

Combined data between field results and vegetation index value then analyzed into biomass using allometric predefined. Allometric calculation of forest cover is a formula to determine the amount of biomass contained in plants [8], [9], [3].

- Stem Biomass allometric

$$\text{WS} = 0,0912 \cdot \text{D}2,22 \quad (2)$$
- Trunk Biomass allometric

$$\text{WT} = 0,0162 \cdot \text{D}2,43 \quad (3)$$
- Leaf Biomass allometric

$$\text{WL} = 0,269 \cdot \text{D}1,7828 \quad (4)$$

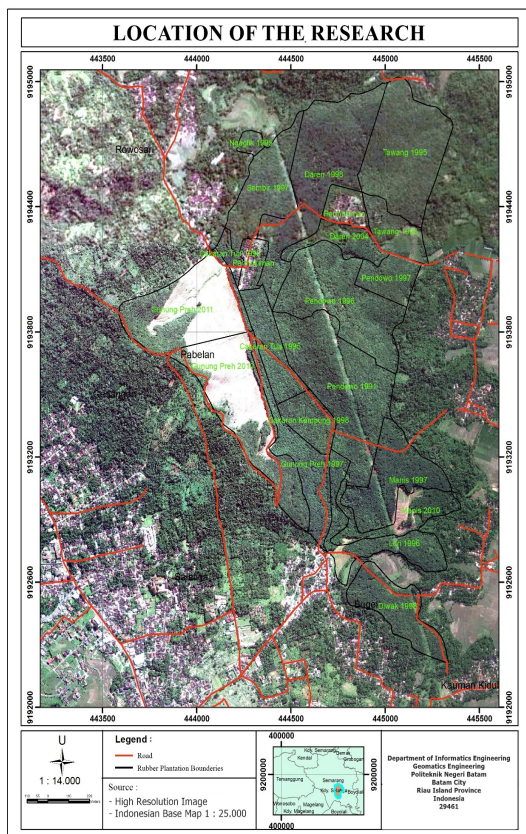


Fig. 3. Location of the study area result and discussion

III. RESULTS AND DISCUSSION

A. Denoising Results Using Raven Pro 1.5

The data used in this image is data level 1B, which the data has been corrected geometric systematically so that the raw data obtained already has reference coordinates of the general location. Systematic geometric correction just put the data into the location of the image the same as the position of the field in General, but there is still a shift from that data against the location details of the research. To adjust the location of the position, the non – systematic geometric correction performed by using three order polynomial algorithm, this is because the physiographic condition of regions mostly hilly and mountainous.

Non – systematic geometric correction use ground control points to tie the data to be corrected using data that already has its own coordinate reference. Basic retrieval point of the belt using the basic map (RBI) scale 1:25,000 in area studies. The retrieval ground control points observe the same appearance between the data that will be corrected with the data used for reference. Geometric correction is carried out to produce the data location of irregularities against the original location on the field. Distortion of position written into RMS magnitude error. The results of geometric correction produce deviation of 0.048404, so that it can be said that the image has a shift by 0.048404 pixels in each pixel, or it could be said that image data is experiencing a shift by 0.68404 meters calculated from the amount of spatial resolution imagery used.

Besides geometric correction, another correction used is radiometric correction. Radiometric correction serves to

restore the spectral values of the image to match the spectral reflection of the objects on the Earth's surface, including in eliminating the interference present in the atmosphere. There are two kinds of processes performed in this radiometric correction conversion of pixel values to at-sensor radiance and conversion at-sensor radiance to at-sensor reflectance. Both of these processes is a process that related to the sensitivity of the sensor that record the electromagnetic energy reflected from the surface of the earth. Converting digital value number (DN) to the reflectance values pass through multiple lifetimes. The first stage is to convert DN values into spectral radiance energy unit. Units of this radiance values are generally $W / m^2 \text{ str } \mu m$, but some are using the unit $\mu W / cm^2 \text{ str } nm$.

The calculations were obtained yield a formula which is then inserted into the image. Sunlight correction calculation result is a formulation reflectance at-sensor used to correct each band on ASTER imagery is

$$(3.142857143 * b_1 * 1.01695864) / (1845.99 * 0.83147)$$

for canal 1

$$(3.142857143 * b_2 * 1.01695864) / (1555.74 * 0.83147)$$

for canal 2, and

$$(3.142857143 * b_3 * 1.01695864) / (1119.47 * 0.83147)$$

for canal 3.

The results of the radiometric correction could then be used for further processing because after going through this process of radiometric correction, spectral reflection contained in the image data used already represent objects on the surface of the earth.

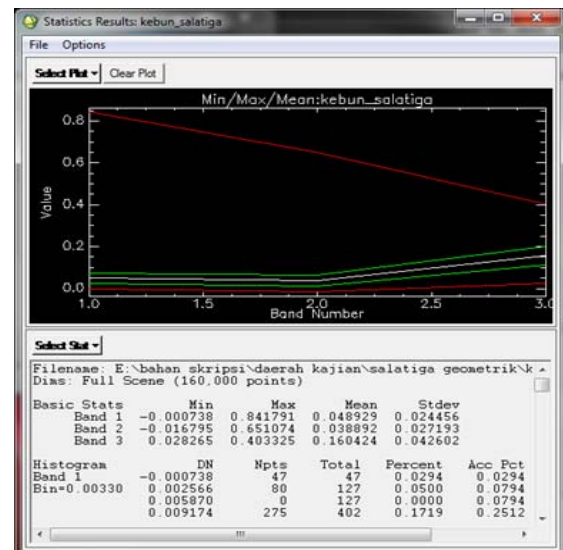


Fig. 4. Image data statistical corrections

Data extraction were done first is land cover analysis using multispectral classification. Multispectral classification is intended to facilitate the grouping classes of objects contained in the image basis on similarity retrieval of spectral values. Classification used is supervised classification with maximum likelihood method where this method classifies pixel values based on the magnitude of pixel probability to get into a particular class. Land cover classes are made is to distinguish between vegetation and non-vegetation. Vegetation objects on ASTER image looks

really stands out with a combination of 321 canal with a predominance of red color. Input in the maximum likelihood classification is the training area as basis for multispectral classification. The training area is the ROI (Region of Interest) taken at each object in an image coverage.

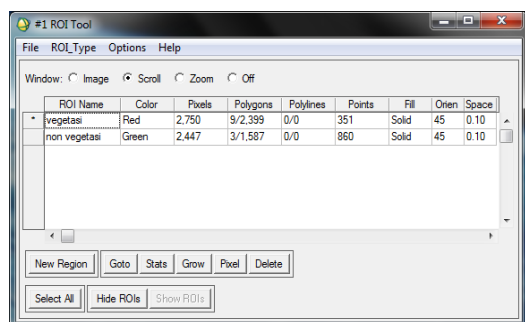


Fig 5. Training area on Region of Interest (ROI)

The same spectral values in the image will be classified into one class of objects equal to the value taken in the ROI. More samples that we put in the sample separation area in the object, more detail of the information we will produce in the classification.

Image Interpretation	Field Survey		Amount	Omisi	Komisi
	Vegetation	Non-Vegetation			
Vegetation	12	2	14	75%	85%
Non Vegetation	4	15	19	88%	78%
Amount	16	17	33	Total Accuracy : 82 %	

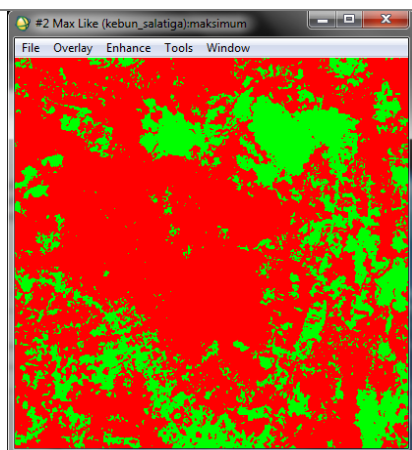


Fig 6. Maximum likelihood classification result of land cover

The multispectral classification is used to separated vegetation and non-vegetation objects. Red color indicates

the vegetation area based on the value of reflect spectral while green color is non vegetation area.

Land cover classification level is then tested using a matrix table error accuracy to compare the correctness of the classification results with the original object on the ground.

Table 1. Error Matrices Test Accuracy of Land Cover Classification

The test results accuracy producing land cover classification accuracy of 82%. This means that the classification results represent 82% of the original appearance of the existing field.

The separation of vegetation and non-vegetation objects are then used as a basis in determining the rubber plantation plot boundaries. This plantation plot delimitation also uses data support high-resolution images to produce more valid garden plot.

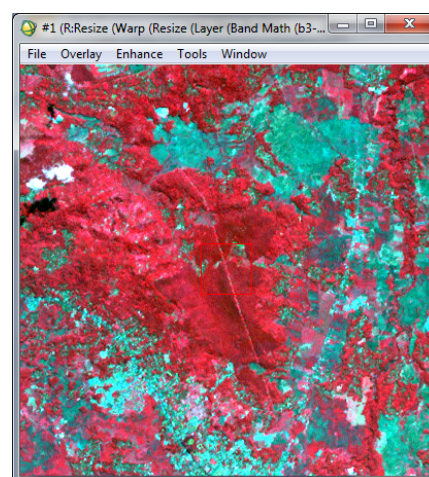


Fig 7. Rubber plantation plot boundaries of ASTER image data

Scheme of making the content of biomass stock's value is by dividing the field sample into two (2) sections that the sample used to create the model and the sample used to test accuracy. The division of the sample was selected based on the proximity to the sample used to create the model with the assumption that the object there closeness has some characteristics with the similarities of the objects that are further apart.

Biomass data derived from remote sensing image by the amount of canopy cover. Vegetation canopy is the first interaction between the waves of the sensor remote sensing of objects on the surface of the earth. The first derivative of ASTER image in knowing the amount of above ground biomass content is by knowing the width of canopy cover on that image. The amount of canopy cover searched using transformation vegetation index (MSAVI) and data width canopy cover is taken out of the field. Both data is then performed a simple linear analysis to enter into the spacious canopy width of image data is used.

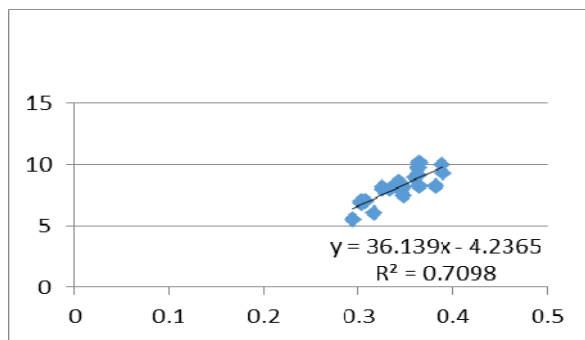


Fig 8. Two-dimensional matrix the relations between width canopy and MSAVI

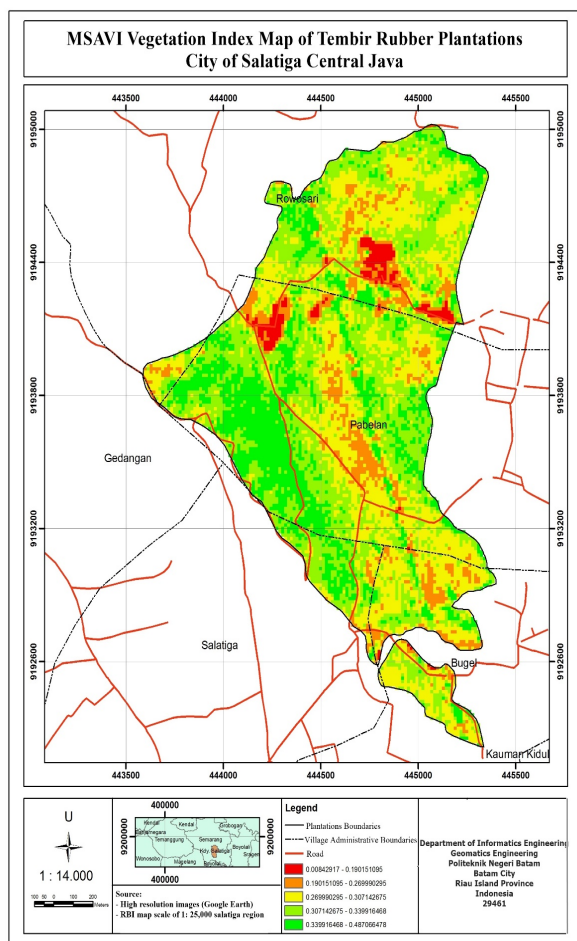


Figure 9. The MSAVI Transformation Index

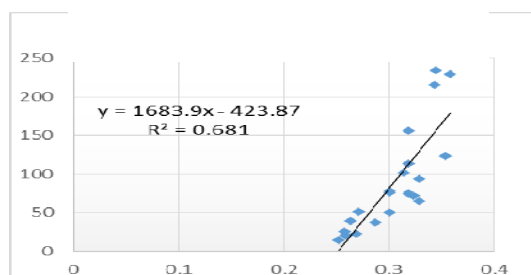


Fig 10. Two-dimensional matrix the relations between width Biomass and MSAVI

The value of $y = 1683.9x - 423.87$ are then added to the image data used to produce a new image data worth vegetation density.

The value of biomass is calculated using allometric calculations with data taken during the field work. Above Ground biomass were targets of this research are Stem Trunk and Leaf biomass in which the total number all three would generate biomass content in the surface of the study area. The relationship between the vegetation index and the magnitude of biomass is known using the regression correlation equation.

Above ground biomass information based on calculation result by using brown alometry showed that the amount of biomass content in the research location was 87,522.25 TonC

Accuracy test is performed to determine the magnitude of errors resulting from field data that has been made modeling. This accuracy test is performed on variables directly related to variables from remote sensing data to ascertain whether information derived from remote sensing data can be used. The result of the accuracy test value Y1 with the MSAVI vegetation index value yields the Standard Error (SE) value of 1,046. The value of SE is arguably very small for the error rate of the data used. This value indicates that at each value generated in the study, only a shift of 1,046 on each model of transformation made. So when viewed from the value, then the data of the biomass produced ranges greater or less than the value of each SE generated.

Table 2. Accuracy test table

No	MSAVI	Y1'	Y1-Y1'
1	0.231	76.049158	1.412608
2	0.232	73.3000963	0.098374
3	0.256	72.522481	0.779249
4	0.261	69.2242861	0.015679
5	0.266	68.3163406	1.065849
6	0.28	64.5509944	0.473806
7	0.28	64.2085138	0.443285
8	0.289	62.0939629	0.416499
9	0.289	61.7199508	0.137805
10	0.29	61.1661283	0.922826
11	0.3	68.331217	4.519846
12	0.304	67.4910238	0.67141
13	0.305	89.7195	1331.567
RMS e (sum/n-2)			1.046768

IV. CONCLUSION

Above ground carbon stock estimation by using remote sensing data in this study yielded the amount of surface biomass content in the study area of 87.522.25 TonC with the level of accuracy of the resulting data of 1.04 which mean the data of the biomass produced ranges greater or less than the value of each SE generated

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